

Jennaya Horne

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EDUCATION

University of San Diego

Bachelor of Science in Computer Science

San Diego, CA

May 2026

TECHNICAL SKILLS

Languages: C, C++, Python, Java

Embedded Systems: STM32F407, ARM-based Microcontrollers, FreeRTOS, Real-Time Systems, Interrupt Handling, Task Scheduling, Memory Management

Systems & Networking: Linux, Socket Programming, Client-Server Architecture, Multithreaded Programming, Network Protocols

Tools: STM32CubeIDE, Git/GitHub, VS Code, Jupyter Notebook

Software Engineering: Object-Oriented Programming, Software Testing, Debugging, Verification & Validation, SDLC, Technical Documentation, Version Control

EXPERIENCE

Software Engineering Capstone

Aug 2025 – May 2026

Becton Dickinson (BD) Senior Capstone

San Diego, CA

- Developed a Python-based software pipeline to preprocess and analyze inrush current signals from BD Alaris infusion pumps for early detection of hardware degradation.
- Built and validated classification workflows to distinguish healthy and degraded device behavior, supporting predictive maintenance for a safety-critical medical device.
- Evaluated system and memory constraints of the embedded pump platform and helped adapt the software approach to meet practical device limitations.
- Collaborated with biomedical engineering teammates on system testing, troubleshooting, validation, and technical documentation throughout the development lifecycle.

Host / Busser

Aug 2023 – Present

Blanco Cocina + Cantina

San Diego, CA

- Collaborated with team members in a fast-paced environment, balancing multiple priorities while communicating effectively and resolving issues as they arose.

PROJECTS

Embedded Real-Time Systems Labs | C, STM32F407, FreeRTOS

- Developed embedded software in C for an ARM-based STM32F407 microcontroller using STM32CubeIDE and FreeRTOS.
- Implemented real-time tasks involving GPIO peripherals, LED control, button input, interrupts, task scheduling, and memory management.
- Applied real-time scheduling and resource management concepts to coordinate concurrent tasks and maintain predictable system behavior.
- Tested and debugged firmware and hardware-software interactions to verify expected embedded system functionality.

Multithreaded Client-Server Networking System | C++, Socket Programming

- Developed a multithreaded client-server application in C++ using socket programming to support concurrent client connections and network communication.
- Implemented synchronization and concurrent request handling to support reliable communication between distributed software components.
- Designed and executed tests to validate protocol behavior, troubleshoot communication failures, and ensure reliable end-to-end functionality.

Maximum Subsequence Sum Analysis | Java

- Designed and implemented multiple Java solutions for the Maximum Subsequence Sum problem while evaluating algorithmic and performance tradeoffs.
- Benchmarked execution times and validated correctness through systematic testing and complexity analysis, improving performance from $O(n^3)$ to $O(n)$.